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SUPERSEDING
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MILITARY SPECIFICATION

DUCT ASSEMBLIES, GROUND, CONDITIONED AIR, INSULATED, FLEXIBLE

* This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers insulated, flexible air duct assemblies for portable ground support air conditioners and heaters.

1.2 Classification. Duct assemblies shall be of any length and of the following diameters, as specified (see 6.2):

Size	Size Code
16 inch	A
12 inch	B
8 inch	C
5 inch	D

*1.3 Part numbering. Part number shall be in accordance with 6.3.

2. APPLICABLE DOCUMENTS.

*2.1 Government documents.

*2.1.1 Specifications standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 Box, Fiberboard

MILITARY

MIL-P-116 Preservation, Methods Of

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Slection Of
MIL-STD-808	Finishes, Protective, And Codes, For Finishing Schemes For Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MS33561	Connection, Aircraft Ground Air Conditioning, 5-Inch, Minimum Requirements

*2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

AIR FORCE

64C1004	Connector-Air Duct Hose, Ground Cooling, Male
64C1005	Connector-Air Duct Hose, Ground Cooling, Female
64C1006	Packing, Preformed, Air Duct Hose, Ground Cooling

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME 19.5:4	Power Test Code Supplement On Instruments And Apparatus

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

AMERICAN SOCIETY FOR TESTING MATERIALS

*	ASTM D430-73	Rubber Deterioration, Dynamic Fatigue, Test For
*	ASTM D471-79	Rubber Property, Effect Of Liquids

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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*2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing.

3.2 Components. The duct assembly shall consist of the following major components:

Duct	(see 3.8.1)
Connectors	(see 3.8.2)
Stiffener core	(see 3.8.3)
Storage device	(see 3.8.4)

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Protective treatment. When materials are used in the construction of the duct assembly that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.4.2 Fire resistance. All exterior materials used in the construction of the duct assembly shall be fire resistant.

3.5 Design. The duct assembly shall be designed for outdoor use in all types of weather including damp, salt-laden air for extended periods of time.

3.6 Construction. The duct assembly shall be so constructed that it may be handled in any direction with a minimum crimping of fabric. The duct assembly shall include a continuous, metal, helical stiffener core to prevent transverse collapsing and excessive area reduction during sharp bends.

3.6.1 Airflow. The internal surface of the duct assembly and the built-in storage device specified in 3.8.4 shall be so constructed that air friction losses will not exceed the following values:

* Diameter (Inches)	Airflow Cubic Feet Per Minute (CFM)	Air Friction Loss (Inches H_2O)
16	3,500	0.15 device + 0.025 per ft of duct
12	2,000	0.18 device + 0.025 per ft of duct
8	875	0.20 device + 0.025 per ft of duct
5	375	0.20 device + 0.035 per ft of duct

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3.6.2 Retractability. The duct assembly shall be so constructed that it may be easily retracted to 25 percent of its full length and compactly stored.

3.7 Performance. The duct assembly shall be capable of satisfactorily delivering conditioned air from portable ground support heating and air conditioning equipment with a maximum air leakage through the duct assembly wall of 0.2 percent of the specified flow rate against a static pressure of 3 pounds per square inch gage (psig) for each duct assembly.

3.7.1 Environmental conditions. The duct assembly shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Ambient storage temperatures ranging from -80° to +160° Fahrenheit (F).

b. Exposure to salt-fog atmosphere.

c. Fungus growth as encountered in tropical climates.

d. Radiant energy as found under natural conditions.

e. Rain.

3.7.2 Operating temperature range. The duct assembly shall be capable of withstanding internal temperatures to 250°F without damage when subjected to ambient temperatures ranging from -65° to +50°F. The duct assembly shall be sufficiently flexible to meet the retractability requirements specified herein at a normal ambient temperature of +70°F, and after a 24-hour exposure to an ambient temperature of -65°F.

3.7.3 Pressures.

3.7.3.1 Internal operating pressure. The duct assembly shall be capable of withstanding an internal operating pressure of 3 psig.

3.7.3.2 Proof pressure. The duct assembly shall be capable of withstanding a proof pressure range from a positive 6 psig to a negative 5 inches of water without permanent damage.

3.7.3.3 Tensile pull. The duct assembly shall be capable of withstanding a tensile pull of 250 pounds without evidence of rupture, separation of the duct from the coupling, or other damage.

3.8 Details of components.

3.8.1 Duct.

3.8.1.1 Fabric. The exterior fabric shall be a strong, nonpermeable material that will meet the requirements and tests specified herein. The fabric shall be highly resistant to abrasion.

3.8.1.1.1 Reparability. When torn or punctured, the duct fabric shall be repairable by stitching or cementing of patches. The stiffener core and carrier shall be so constructed that, after being seriously deformed, they can be manually repaired to approximate their original forms.

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3.8.1.2 Insulation. Sufficient insulating material, of a type that will not wad or powder during the life of the duct, shall be utilized to provide a maximum overall coefficient of heat transfer (U) of 0.75 BTU/hr-ft^2-°F for each size duct.

3.8.1.3 Scuff strip. The duct shall be protected by means of a scuff strip or shall be so constructed that the basic fabric will be protected from excessive abrasion when dragged over level surfaces at speeds from 4 to 5 miles per hour (mph). If stitching is used, materials shall be capable of meeting the requirements and tests specified herein. Stitching shall be of the lock type to prevent raveling when a stitch is broken.

3.8.2 Connectors. When specified (see 6.2), each section of duct assembly shall incorporate a male connector (with seal) at one end and a female connector at the other end. The 5-inch female connector shall conform to MS33561. The 5-inch male connector shall be a mating movable claw type. The 8- and 16-inch female connectors shall conform to Drawing 64C1005. The 8- and 16-inch male connectors shall conform to Drawing 64C1004. The 8- and 16-inch seals shall conform to Drawing 64C1006. The 12-inch male connector, female connector and seal shall conform to Drawing 64C1004, 64C1005, and 64C1006, respectively, except dimensions shall be changed as required to meet the 12-inch duct requirements.

3.8.2.1 The 5-inch male and female connectors shall be constructed to withstand repeated drops onto concrete from a height of 8 feet while attached to the duct assembly.

3.8.3 Stiffener core. The stiffener core shall be so constructed that it will retain an approximate circular cross-sectional area of the duct assembly. The stiffener core shall continue around the duct assembly and square off at both ends. The attaching cuff on the connector shall be as narrow as practicable to permit securing the connector close to the squared-off stiffener core at the ends of the duct. When the duct assembly is released from the carrier, the core shall not be of such rigidity as to interfere excessively with retraction of the duct to the storage position. The stiffener core and attaching cuff shall not work loose under repeated use. Separation of the inner liner and outer cover which allows the stiffener core to move from its design location, or separation of the attaching cuff during any of the tests specified in 4.6 shall be cause for rejection.

3.8.3.1 Pitch. To prevent excess area reduction during 90-degree bends, nominal pitch of the stiffener core shall be held to the minimum consistent with retractability requirements and shall not exceed 95 percent of the nominal diameter of the duct assembly.

3.8.4 Storage device. A built-in storage device shall be internally located in each duct assembly section to act as a guide in retracting the duct assembly and to firmly retain it in the storage position. The storage device shall not damage the fabric during extension or retraction of the duct assembly. The carrier portion shall be firmly attached to the male coupling. All welds on the storage device shall be of the fusion type. In the event that carrier cleats are riveted to the coupling ends, two rivets shall be used on each end of the cleat.

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3.9 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-100 shall govern the manufacturer's part number and changes thereto.

3.10 Dimensions. Duct assembly lengths shall be held to specified (see 6.2) lengths +1 foot -0 feet. When connectors are incorporated, the overall length shall include the couplings. The inside diameters shall be as follows:

16 inch: 16 +1/4, -0 inches

12 inch: 12 +1/4, -0 inches

8 inch: 8 +1/4, -0 inches

5 inch: 5 +1/4, -0 inches

3.10.1 When retracted for storage, the overall length of the duct assembly including end connectors shall not exceed 25 percent of the specified (see 6.2) extended length. The outside diameters of the assemblies shall not exceed the following:

16 inch: 20 inches

12 inch: 16 inches

8 inch: 12 inches

5 inch: 9 inches

3.11 Weight. The weight of the complete duct assemblies, including end connectors, shall not exceed the following:

16 inch: 13.6 pounds +3.9 pounds per foot

12 inch: 10.4 pounds +2.5 pounds per foot

8 inch: 6.7 pounds +1.6 pounds per foot

5 inch: 4.2 pounds +1.0 pounds per foot

*3.12 Colors. The exterior surface of the ducts shall be dark green color 24052 in accordance with MIL-STD-808.

3.13 Cleaning, painting, plating, chemical films, and chemical treatments. All cleaning, painting, plating, chemical films, and chemical treatments shall be in accordance with MIL-STD-808.

3.14 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. Marking shall include manufacturer's name, part number, contract number, cure date, and Federal stock number.

3.15 Workmanship. Workmanship shall be in accordance with high grade commercial practice.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the duct assembly shall be classified as follows:

a. First article testing (see 4.4)

b. Acceptance test (see 4.5)

4.3 Test conditions.

4.3.1 Apparatus. Insofar as practicable, apparatus used for testing shall be of laboratory-precision type and calibrated at properly spaced intervals to include laboratory accuracy. All measurements for the following test conditions shall be measured in accordance wth ASME PTC 19.5:4.

4.3.1.1 Airflow. Airflow shall be measured by means of smooth-approach orifices or thin plate orifices. Apparatus such as pitot tubes, rotometers, et cetera, may be used only when specifically approved by the procuring activity. The rate of airflow shall be expressed in cubic feet per minute.

4.3.1.2 Barometric pressure. Barometric pressure shall be measured by means of a mercurial barometer. The barometric pressure shall be corrected according to the temperature of the scale and mercury and the location of the barometer with regard to the altitude and latitude. Readings obtained by aneroid barometers shall not be acceptable.

4.3.1.3 Temperatures.Temperature shall be measured by means of appropriately located thermocouples. Insofar as practicable, the thermocouples shall be insulated to prevent contact with other metals. Temperatures shall be expressed in degrees Fahrenheit. Test temperatures shall be within $\pm 4^{\circ}$F of those specified herein.

4.3.1.4 Pressure. Insofar as practicable, pressure and pressure differentials shall be measured by means of water manometers having a scale graduated in 0.01 inch of water. Pressures that exceed the range of conventional water manometers shall be measured by properly calibrated bourdon-tube gages. Gages shall be laboratory-precision type and shall have a dial diameter of not less than 4 inches.

4.3.1.5 Atmospheric conditions. Unless otherwise specified, tests shall be conducted at atmospheric pressure (approximately 29.22 inches Hg) and at temperatures ranging from 70° to 80°F.

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4.3.2 Readings. Readings of all pertinent pressures, temperatures, and other data necessary to demonstrate compliance with applicable portions of Section 3 shall be recorded at intervals not exceeding 30 minutes during all tests, unless otherwise specified herein.

4.4 First article testing.

4.4.1 Test samples. Two duct assemblies of each diameter being furnished and samples of all materials used in construction of the duct assemblies shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. Upon completion of the first article tests, a first article test report in accordance with MIL-STD-831 shall be prepared.

4.4.3 First article tests. The first article test samples shall be allocated to the first article tests as follows and tested as described under 4.6:

Sample No. 1	Sample No. 2
Examination of product (4.6.1)	Examination of product (4.6.1)
Airflow tests (4.6.2)	Drop test (4.6.7)
Pressure test (4.6.3)	Expansion test (4.6.8)
Leak test (4.6.3.1)	Fire resistance test (4.6.5)
Tensile load test (4.6.4)	Abrasion resistance test (4.6.6)
High temperature test (4.6.14.1)	Absorption test (4.6.9)
Low temperature test (4.6.14.2)	Tensile strength test (4.6.10)
Salt-fog test (4.6.14.3)	Fabric flex life test (4.6.11)
Sunshine test (4.6.14.5)	Fungus-resistance test (4.6.14.4)
Reparability test (4.6.12)	Bend test (4.6.13)
Rain test (4.6.14.6)	

4.5 Acceptance test. The acceptance test shall consist of the individual test.

4.5.1 Individual test. Each duct assembly shall be subjected to the test specified in 4.6.1.

4.6 Test methods.

4.6.1 Examination of product. The duct assembly shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, proper assembly, dimensions, and marking.

4.6.2 Airflow tests. Airflow tests shall be conducted on full-length duct assemblies with storage racks. The air friction loss for each duct size shall not exceed the values specified in 3.6.1 for the airflows specified. During testing, the duct shall be extended to full length, but shall not be placed under tension. The inlet end of the duct assembly shall be connected to the airflow and static-pressure measuring duct with the outlet of the air duct unrestricted.

4.6.3 Pressure test. One duct assembly of each size shall be subjected to pressure tests. Blank fittings shall be attached to the end connections and shall form airtight seals when subjected to an internal pressure of 6 psig. One fitting shall have provisions for regulating the internal pressure, and the other have provisions for pressure measurement. The duct assembly

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shall be fully extended when pressurized. The end connections shall be anchored to maintain at least 90 percent extended duct length when negative pressure is applied. The duct assembly shall withstand the pressure specified in 3.7.3.2 without rupture or other damage.

4.6.3.1 Leak test. One duct assembly of each size shall be subjected to a leak test to demonstrate compliance with 3.7. The test shall consist of a 30-minute run with readings recorded at 10-minute intervals.

4.6.4 Tensile load test. One duct assembly of each size shall be subjected to a tensile load test. In testing the duct assembly, blanks shall be secured to the end connectors and the duct subjected to a tensile load of 250 pounds. All forces shall be applied to the center of the blanks. No rupture or other damage or separation of the duct assembly and connector shall occur as a result of this test.

4.6.5 Fire resistance test. Samples of materials used in the fabrication of the exterior surface of the duct assembly shall be tested for fire resistance in accordance with FED-STD-191, Method 5904.

4.6.5.1 Three 2-3/4- by 12-1/2-inch pieces of each type of material shall be used. Where possible, one-half of each set shall be cut with the long dimension in the direction of the warp and the other half with the long dimension in the direction of the fill. Pieces of stitching material or nonmetallic scuff strips shall be 12-1/2 inches long.

4.6.5.2 The duration of flame and glow in the specimens shall be timed. After complete extinction of all flame and glow, the char length shall be measured. No specimen shall support flame more than 3 seconds nor glow more than 5 seconds after the burner is withdrawn. Average char length of any set of test specimens shall not exceed 5 inches. Char length shall be the distance from the exposed end of the specimen in which any portion of the material is destroyed or transformed from its normal state into an incoherent, softened, or ashlike substance.

4.6.6 Abrasion resistance test. Abrasion resistance of the duct fabric and the duct assemblies shall be determined as follows:

4.6.6.1 Pieces of new duct assembly exterior fabric and the exterior fabric removed from ducts which have undergone the sample No. 1 testing sequence shall be tested in accordance with FED-STD-191, Method 5306. The Taber abraser shall employ H-22 abraser wheels under a load of 1,000 grams. Tests shall be conducted on the side of the material which would be the external side of the duct. At the end of 1,000 wear cycles, the fabric coating shall not be worn to such extent that the basic fabric is torn by the abraser wheels. Magnification shall be used to aid in determining the extent of wear.

4.6.6.2 A 4-inch diameter disc with a swatch of the scuff strip material attached shall be subjected to an abrasion resistance test simulating wear that would occur over a clean, dry, brush-finish concrete surface. The test shall be conducted using a Taber abraser equipped with H-22 calibrade wheels with an applied load of 1,000 grams. The specimen shall be rotated for a total of 10,000 cycles with an inspection of the swatch being made after every 1,000 wear cycles to note any wear. Any wear through the scuff strip to the backing material so as to damage the duct shall be cause for rejection.

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4.6.7 Drop test. One 5-inch duct assembly, with end connectors and built-in storage device, shall be subjected to a drop test. Each end of the duct assembly shall be raised to a height of 8 feet and dropped onto a concrete slab 20 consecutive times. Any damage as a result of this test which would cause difficult connection or leakage shall be cause for rejection.

4.6.8 Expansion test. One duct assembly of each size shall be tested to insure compliance with the expansion requirements of 3.8.3. The duct assembly shall be retracted and extended 20 times. Any evidence of the helical stiffener core failing to extend the duct assembly to at least 85 percent of its full length shall be cause for rejection.

*4.6.9 Absorption test. All materials on the exterior of the duct shall be tested for resistance to gasoline, oil and water in accordance with ASTM Standard D471-79. Pieces of duct fabric, scuff-strip fabric, and stitching shall be soaked in 91-octane gasoline, SAE No. 10 oil and water. At the end of 48 hours, the material shall be examined to determine that the increase in thickness, weight and volume does not exceed 10, 15 and 20 percent respectively. All other materials shall be tested in the same manner, but for resistance to water only.

*4.6.10 Tensile strength test. Breaking strength of non-metallic materials used in the duct and tear strength of the duct fabric shall be determined by the cut-strip and falling pendulum methods, respectively, as specified in FED-STD-191, Methods 5102 and 5132. Pieces of material cut from ducts that have been subjected to the sample No. 1 testing sequence shall exhibit at least 75 percent of the strength of the same material when new.

*4.6.11 Fabric flex life test. Flex life of the duct assembly fabric shall be tested by subjecting six 1- by 6-inch pieces of each material to 10,000 band flex cycles in a DeMattia Flexing Machine, or equal, in accordance with ASTM Standard D430-73. Half of the samples shall be cut in the direction of the warp, and the other half in the direction of the fill. The groove described for 1/4-inch thick samples shall not be used. Following the band flexing, the duct material shall be checked for breaking strength in accordance with 4.6.10 and shall exhibit 75 percent of the strength of unflexed fabric.

4.6.12 Reparability test. Conformance to 3.8.1.1.1 shall be determined by demonstration of fabric patching methods. Reparability of the stiffener core shall be determined by flattening samples of each size duct to one-third of their original diameter and then returning the duct to the approximate original shape without the use of special tools. After repair, the duct assembly shall be subjected to the airflow tests specified in 4.6.2 and shall not show an increase in air friction loss in excess of 0.05 inch of water. In addition, compliance with the requirements of 3.10.1 shall be demonstrated.

4.6.13 Bend test. A section of each size duct shall be submitted to 500 90 degree bends over a drum diameter twice the diameter size of the duct. No damage shall be noted, particularly to the scuff strip, stiffener core or attaching cuff.

4.6.14 Environmental test.

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4.6.14.1 High temperature test. One complete duct assembly of each size shall be subjected to a high-temperature test. The duct assembly shall be retracted to a storage position, placed in a test chamber, and the temperature raised to and maintained at 160°F for 24 hours. Following this exposure, the duct assembly shall be fully extended and then retracted to its storage position 20 times. Following this portion of the test, the duct assembly shall be attached to a source of air having a temperature of 200 ±5°F and shall be arranged to be supported only by the floor of the test space with the complete length of the duct assembly in contact with the floor. The airflow through each size duct assembly more than 90 percent recirculated air. The ambient air temperature of the test chamber, measured at any point 1 foot from the periphery of the duct assembly, shall be 50°F. If air moving devices are used in testing, baffles shall be placed around the duct assembly to prevent air from blowing on the duct assembly. Each duct assembly shall be tested for 150 hours. At the end of the 150-hour test, the temperature of the air passing through the duct assembly shall be increased to 250°F and the duct assembly tested at this temperature for 2 hours with all other test conditions remaining the same. At the conclusion of the test, the duct assembly shall exhibit no evidence of scorching nor other harmful effects. During this test, the requirements of 3.8.1.2 shall be demonstrated.

4.6.14.2 Low temperature test. Following the exposure specified in 4.6.14.1, one complete duct assembly of each size shall be placed in a test chamber in the retracted position and cooled to and maintained at -80°F for 24 hours. Damage to the duct assembly as a result of this exposure shall be cause for rejection. The temperature shall then be raised to and maintained at -65°F for the remainder of the test. After a 24-hour soak in the retracted position, the duct assembly shall be fully extended and retracted to storage position 30 times. The duct assembly shall then be given another 24-hour soak in the extended and retracted position to storage 20 times. The flexibility and retractability of the duct assembly shall be observed and recorded. The time required to retract the duct assembly to storage position shall not exceed 1 minute. At the conclusion of this test, the materials shall exhibit no harmful effects due to flexing and exposure. The duct assembly shall then undergo the tests specified in 4.6.3 and 4.6.4.

4.6.14.3 Salt-fog test. One duct assembly of each size shall be subjected to salt-fog in accordance with Method 509.1, procedure I, of MIL-STD-810 for 50 hours. Upon completion of the test, there shall be no evidence of serious corrosion of metallic parts nor other damage that might affect subsequent operation.

4.6.14.4 Fungus-resistance test. If fungus nutrient materials are used in the duct assembly, material samples from one duct assembly of each size shall be subjected to the fungus-resistance test specified in Method 508.1, procedure II, of MIL-STD-810.

4.6.14.5 Sunshine test. One duct assembly of each size shall be subjected to sunshine in accordance with Method 505.1 of MIL-STD-810.

4.6.14.6 Rain test. One duct assembly of each size shall be subjected to simulated rain per method 506 of MIL-STD-810.

4.6.15 Retractability test. One duct assembly of each size shall be tested to insure compliance with 3.6.2, 3.8.3, and 3.8.4. The duct assembly shall be extended and retracted 20 times with a maximum effort of 25 pounds force. A

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force greater than 25 pounds required by the individual to retract the duct assembly to at least 25 percent of its full extended length shall be cause for rejection.

4.6.16 Insulation shift test. One duct assembly of each size shall be tested to insure compliance with 3.8.1.2. A 3-foot section of each duct size shall be mounted in an extended vertical position on a vibration stand and subjected to 10,000 vibration cycles with a stand deflection of 1/8 inch. Following the vibration test, the duct section shall be checked for an insulation shift and shall show not more than a 1/4 inch shift per foot of duct.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING.

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Packaging shall be in accordance with Method III of MIL-P-116. Each section shall be retracted as specified in 3.10.1 for packaging in the unit container.

5.1.2 Level C. Packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Duct assemblies, packaged as specified in 5.1.1, shall be packed one each in domestic-type container conforming to PPP-B-636.

5.2.2 Level B. Duct assemblies, packaged as specified in 5.1.1, shall be packed one each in domestic-type container conforming to PPP-B-636.

5.2.3 Level C. Duct assemblies shall be packed for shipment in commercial exterior shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery.

5.3 Shipment marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended Use. The duct assemblies covered by this specification are intended for use in conjunction with ground support heating and air conditioning equipment to deliver the conditioned air from the ground equipment to aircraft and missile electronics equipment and personnel compartments, portable maintance shelters and hangars, and similar enclosures.

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*6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Use Definitive Part Number (see 6.3).

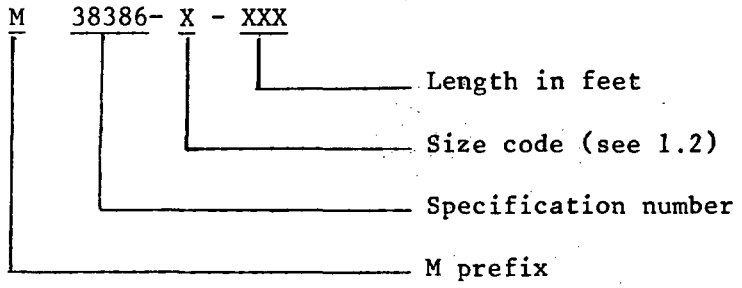
c. When duct connectors are to be furnished (see 3.8.2).

d. Required level of packaging and packing (see Section 5).

e. Special shipment marking (see 3.14 and 5.3).

*6.3 Part number. Part number shall be in accordance with the following:

Example:



6.4 First article. When required a first article inspection shall be conducted per 4.4. First article is required where bidder has not delivered satisfactory items in the 24 months preceding solicitation of bids. The first article test report will be forwarded to the procuring activity for evaluation.

*6.5 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

Project 4720-F637

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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